Bayer Leverkusen 2023/24 Pressing analysis

Bundesliga event data | All 34 league matches

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This project looks at Leverkusen's pressing during their unbeaten Bundesliga season. The main aim was to measure how often pressure was followed by a quick regain and whether those regains produced more attacking value than other possessions starting in similar areas.

Research question

Did quick pressured regains create more danger than Leverkusen's other open-play possessions beginning in comparable areas?

Season summary

Metric	Result
Fully observed pressured possessions	1,702
Regained within five seconds	538 (31.6%)
Quick regains producing a shot within 15 seconds	46 (8.6%)
Goals from those shot sequences	4

Main findings

- Penalty-area reach was almost unchanged after location matching: 15.3% after quick regains versus a 14.9% expectation.
- Shot probability was higher (6.7% versus 5.0%), but the bootstrap interval included zero.
- The point estimate for xG rose from 0.65 to 1.22 per 100 possessions, although this result was also uncertain.
- The spatial analysis suggested a possible wide pressing pattern in the middle third.

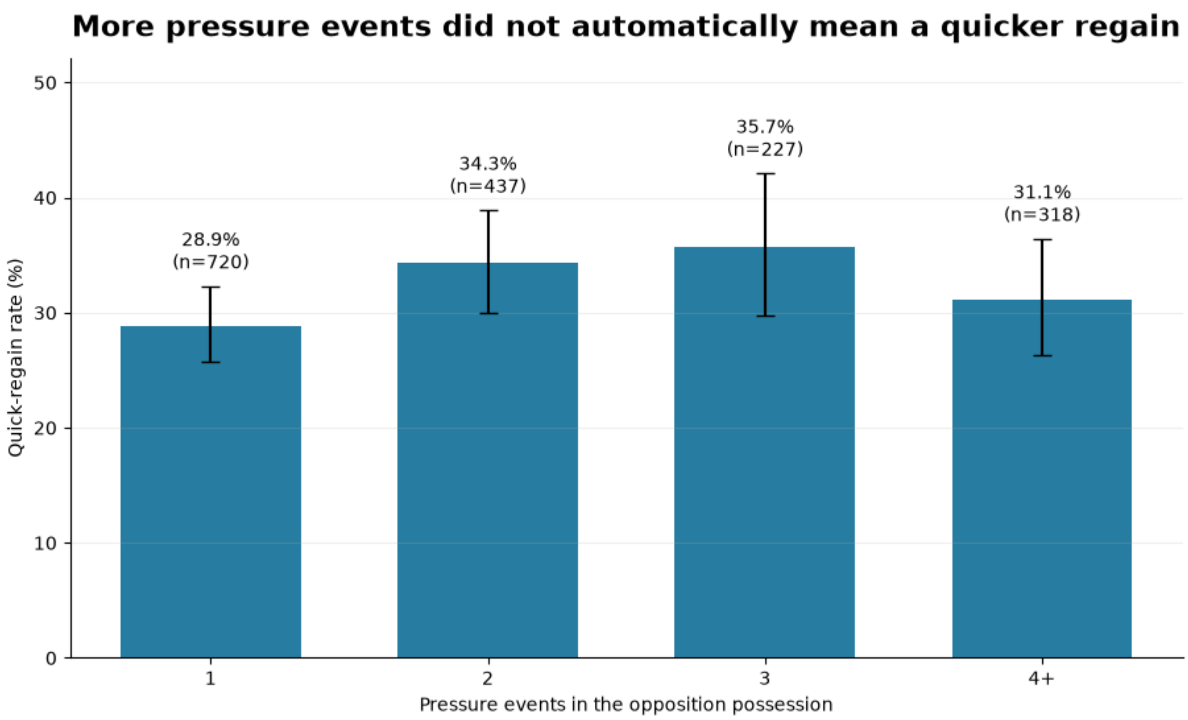
Data source: StatsBomb Open Data accessed through mplsoccer.

1. Method and KPI definitions

One analytical row represents one pressured opposition possession, rather than one pressure event. This avoids counting a possession several times when multiple Leverkusen players apply pressure.

Term	Definition
Pressure sequence	An opposition possession containing at least one Leverkusen pressure event. The final pressure location represents the sequence.
Quick regain	The next possession belongs to Leverkusen within five seconds and the opponent does not shoot before possession changes.
Attacking value	Shots, xG and penalty-area entries recorded during the first 15 seconds of the regained possession.
Matched baseline	Other open-play Leverkusen possessions starting in the same third and lateral channel provide the expected outcome rate.

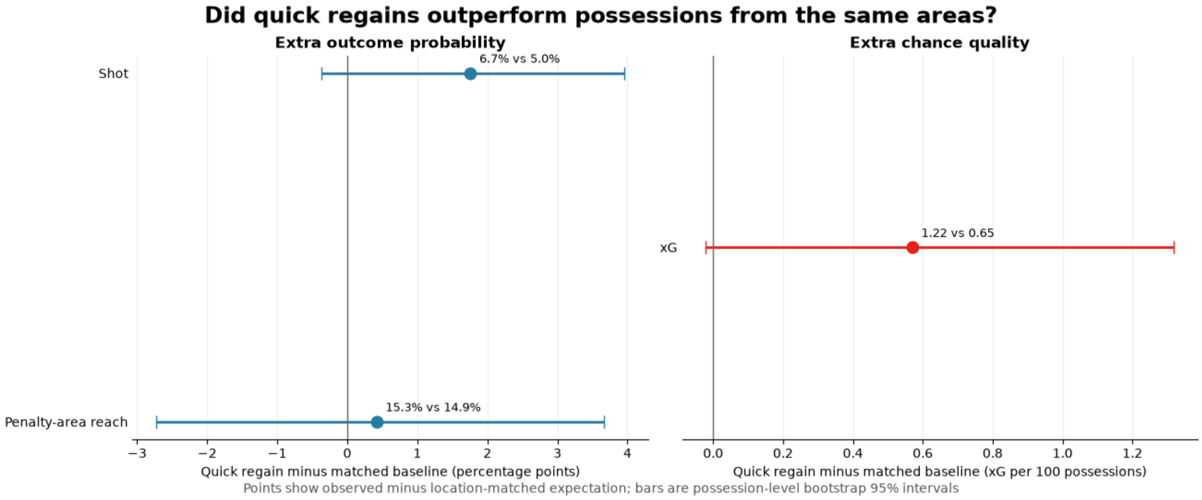
Pressure events within the possession



Possessions containing two or three pressure events had higher raw quick-regain rates than single-pressure possessions. The rate fell for possessions with four or more pressure events, so there was no simple dose-response relationship.

2. Attacking value after quick regains

High regains naturally start closer to goal, so comparing them with all possessions would be misleading. Each quick regain was therefore compared with ordinary open-play possessions from the same coarse starting area. A total of 489 quick regains were retained after matching.



Outcome	Quick regain	Expected	Difference	95% interval
Penalty-area reach	15.3%	14.9%	+0.4 pp	(-2.7, 3.7)
Shot	6.7%	5.0%	+1.8 pp	(-0.4, 4.0)
xG per 100	1.22	0.65	+0.57	(-0.02, 1.32)

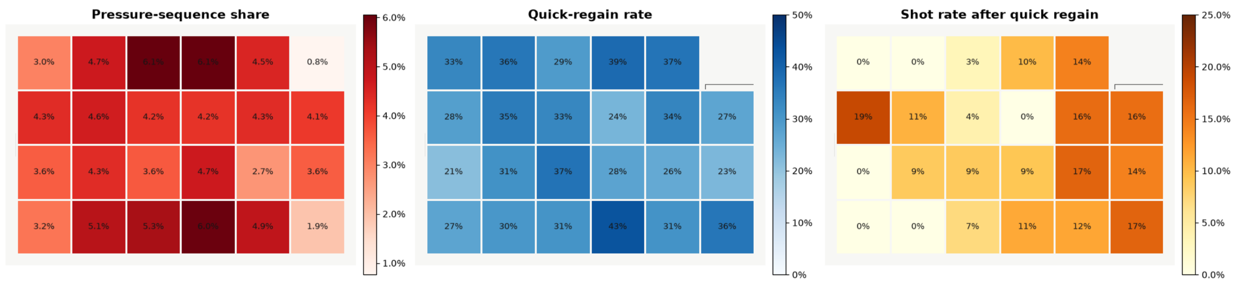
Interpretation

- Quick regains did not improve territory progression beyond what would be expected from their starting locations.
- The shot and xG point estimates were higher, suggesting that the defensive structure may have been less settled after a regain.
- Both uplift intervals included zero, so the evidence is suggestive rather than decisive.

3. Spatial pressure patterns

The heatmaps separate where pressure sequences occurred, where they were most likely to produce a quick regain, and where those regains were most likely to lead to a shot. Low-sample cells are left blank.

Where Leverkusen pressed, regained, and created



Final pressure location · Leverkusen attack from left to right · blank cells are below the stated sample threshold

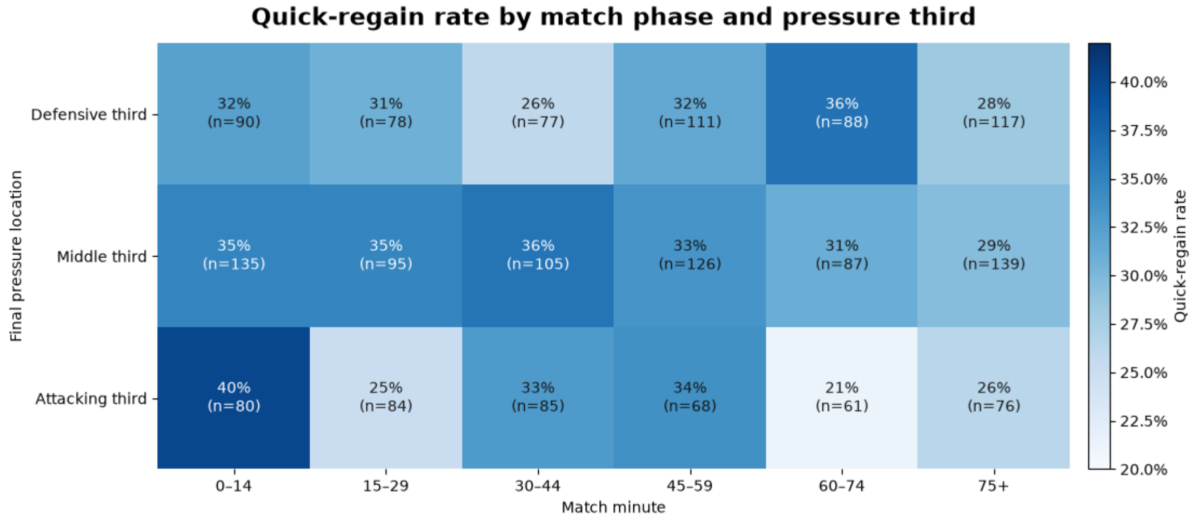
Final pressure location	Possessions	Regain rate	Shot rate per regain
Defensive third	561	30.8%	5.2%
Middle third	687	33.2%	7.9%
Attacking third	454	30.2%	13.9%

Possible wide middle-third pattern

The two outer channels of the middle third cover 16.7% of the pitch but contained 23.4% of final pressure locations. These possessions had a 35.7% quick-regain rate, compared with 30.4% elsewhere. This is consistent with the touchline helping to restrict the opponent, but event locations alone cannot confirm that it was a deliberate pressing trap.

The attacking-third shot rate was higher because these regains started closer to goal. This is why the location-matched comparison on the previous page is used for the main conclusion.

4. Match phase and limitations



Match-phase observations

- Attacking-third pressure produced a 40% quick-regain rate in minutes 0-14 (n=80).
- The same rate was 21% in minutes 60-74 (n=61). This could reflect game state, substitutions, opponent adaptation or reduced intensity.
- Five matches supplied 60% of the season's quick-regain xG, showing that the attacking return was uneven across fixtures.

Limitations

- Event data does not show the full off-ball team shape available from tracking data.
- A quick regain following a recorded pressure does not prove that the pressure caused the turnover.
- The matched baseline controls only a coarse starting location, not score state, opponent strength or player availability.
- Bootstrap intervals treat the matched baseline rates as fixed and should be read as descriptive uncertainty estimates.
- The results describe Leverkusen's 2023/24 Bundesliga season and should not automatically be generalised to other teams or competitions.